

Platform Notes

Vector Indexing

The vector store keeps a dedicated HNSW graph per embedding model so a cosine scan never crosses model boundaries.

Each graph stores points by approximate nearest neighbour links and answers a top-K query in a small beacon scan.

The index applies to a fixed vector dimension because the graph geometry is built for that dimension at build time.

A second embedding model therefore builds its own graph rather than sharing the existing one.

Queue Processing

Jobs reserve a lease that must be renewed before the visibility timeout, otherwise the job is offered back to the queue.

A lease identifies a single worker so two workers never process the same job at the same time.

Failed attempts retry with exponential backoff up to a configured ceiling before the job moves to a dead state.

Successful jobs release the lease and record the outcome so the orchestrator can emit metrics.

Token Budgets

A token budget of 512 tokens keeps most semantic units intact while bounding a single embedding request.

Paragraphs that exceed the budget are hard-split at word boundaries so no single unit is ever dropped.

The budget is measured with a character based estimate that stays within a few percent of a real tokenizer on prose.

Smaller budgets multiply the number of units and raise the embedding cost for the same corpus.

Multi-Tenancy

Every retrieval query is scoped to the requesting tenant at the SQL layer, and a leak across tenants is a release-blocking defect.

Chunks and embeddings carry the organization id on every row so filtering never depends on application memory.

The tenant scoping applies at documents, chunks, embeddings, and ingestion jobs alike.

Cross tenant queries return an empty result set rather than exposing another tenant's rows.

Retrieval Latency

Index lookups complete in tens of milliseconds at small scale because the top-K scan touches only the best-matched nodes.

Latency grows with the number of embedding models in play, since each model scans its own graph.

Worst-case behaviour is a sequential scan over every embedding of every model, which the system avoids by index choice.

The latency budget for a search request is kept under one second end to end at the p99 percentile.

Overlap Strategy

Consecutive sliding windows share a tail of overlap tokens so a claim split across a boundary still matches a single query.

The overlap tail is much smaller than the window itself; typical settings share 40 to 80 tokens.

Overlap raises storage only slightly while it meaningfully improves boundary recall for long passages.

The tail duplicates the shared text across both windows, so the embedding of each window stays contiguous.